

Getting Started with NHANES in R: Data Structure, Data wrangling, and Visualizations

Use NHANES documentation, relative paths, XPT imports, SEQN joins, and beginner ggplot figures in one organized R workflow.

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Workshop agenda: NHANES data structure and R demo

5 min

Workshop goals and NHANES overview

What NHANES measures and why nutrition researchers use it

10 min

Finding and organizing NHANES files

Cycles, documentation, file families, data/raw, and data/processed

15 min

R demo: import and join

`relative_path`, `read_xpt()`, `select()`, `left_join()`

15 min

Visualization

`ggplot` checks and `ggsave()`

15 min

Practice and wrap-up

build one plot; What to check before real analysis



National **H**ealth **a**nd **N**utrition **E**xamination **S**urvey

<https://www.cdc.gov/nchs/nhanes/index.html>



NHANES connects interviews, exams, labs, and diet

NHANES is a CDC/NCHS program that combines interviews, physical examinations, laboratory measures, and dietary information.

The screenshot shows the NHANES website interface. At the top, the CDC logo and text "Centers for Disease Control and Prevention CDC 24/7: Saving Lives. Protecting People™" are visible. Below this is the "National Center for Health Statistics" header. The main content area features the "National Health and Nutrition Examination Survey" title and "NHANES Questionnaires, Datasets, and Related Documentation". A left sidebar contains navigation links: "National Health and Nutrition Examination Survey", "About NHANES", "Latest Data Releases", "Questionnaires, Datasets, and Related Documentation" (which is expanded to show "Survey Methods and Analytic Guidelines", "Data User Agreement", "Search Variables", and "Frequently Asked Questions"), and "Print". The main content area includes a "Print" link, a "Survey Methods" section with a document icon, a "Search Variables" section with a magnifying glass icon, and a link to "Guidelines for High Quality Analyses of NHANES Data".

Data, Documentation, Codebooks

-  Demographics Data
-  Dietary Data
-  Examination Data
-  Laboratory Data
-  Questionnaire Data
-  Limited Access Data

What is an NHANES cycle?

An NHANES cycle is a public-use data-collection period.

- Traditionally released in approximately two-year periods
- Each cycle contains a new cross-sectional sample
- A cycle does **not** follow the same participants over time
- File names identify which release the files belong to
- Match cycle markers when merging files

Reading NHANES File Names:

File stem + cycle marker

- **1999–2000:** DEMO — _A is implied but omitted
- **2001–2002:** DEMO_B
- ...
- **2017–2018:** DEMO_J
- **Special prefix 2017 – March 2020 pre-pandemic:** P_DEMO
- **August 2021–August 2023:** DEMO_L

NHANES Data Categories

Demographics

age, sex, race/ethnicity, income,
survey design variables

Dietary

24-hour recalls, nutrient intake,
supplement use

Examination

Body measures, blood pressure, dental
and physical measures

Laboratory

Biomarkers from blood, urine, and other
specimens

Questionnaires

health history, behaviors, medications,
food security

Limited Access

Restricted geographic or sensitive
variables

NHANES File Families and Units of Record

Family	Typical files	Unit of record / timing
Demographics	DEMO	Person; one record per participant per cycle
Dietary	DR1TOT/DR2TOT; DR1IFF/DR2IFF; supplement files	Person-day for totals; person-food for individual foods; Day 1 and Day 2 recalls; <i>Often requires aggregation or scoring, such as calculating a person-level HEI score.</i>
Examination	BMX, BPX	Person or examination visit; repeated readings may be separate variables
Laboratory	GLU, HDL, TRIGLY, TCHOL	Person/specimen; generally measured at the examination; some use subsamples
Questionnaire	DIQ, PAQ, DBQ	Person; reference period varies by question

Before merging, ask: What does one row represent, who was eligible, and which weight applies?

Why NHANES is useful for nutrition research

NHANES is one of the most widely used public health datasets for nutrition and epidemiology research.

It support projects involving:

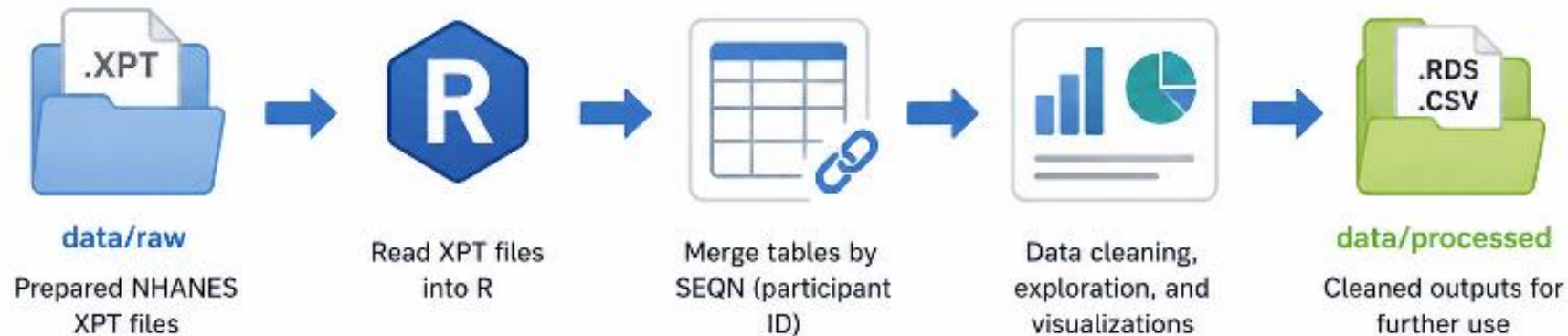
- Dietary intake
- Biomarkers and laboratory measures
- Anthropometrics (height, weight, BMI, etc.)
- Health conditions and risk factors

Always starts with:

- ✓ Survey cycle
- ✓ Codebook
- ✓ Eligibility criteria
- ✓ Missing-value codes
- ✓ Appropriate survey weights



Today's workflow



In today's demo, we use prepared XPT files in **data/raw** and write cleaned outputs to **data/processed**.

The live R code demonstrates the workflow rather than a full survey-weighted analysis.

Resources:

- About NHANES: <https://www.cdc.gov/nchs/nhanes/about/index.html>
- NHANES datasets and documentation: <https://wwwn.cdc.gov/nchs/nhanes/default.aspx>
- Continuous NHANES: <https://wwwn.cdc.gov/nchs/nhanes/continuousnhanes/>
- NHANES tutorials: <https://wwwn.cdc.gov/nchs/nhanes/tutorials/default.aspx>

Treat documentation as part of the dataset

A clean project makes it easy to rerun analysis and explain every recode.

```
NHANES_project/
├── data/
│   ├── raw/           Original, untouched NHANES XPT files
│   └── processed/      Cleaned, merged, or analysis-ready datasets
├── code/              R Markdown code file
├── docs/
│   └── documentation/  Saved CDC codebooks and documentation
├── outputs/
│   └── figures/        Saved plots, tables, and other outputs
```

Before coding

1. Confirm the cycle
2. Save the Data file and Doc File
3. Read eligibility and missing-value notes
4. Identify the join key: SEQN
5. Identify the correct survey weight

Data File Name	Doc File	Data File	Date Published
Demographic Variables and Sample Weights	DEMO_J Doc	DEMO_J Data [XPT - 3.3 MB]	February 2020

RStudio set up for the live R Markdown demo

The image shows the RStudio interface with four main panes highlighted by colored boxes and text annotations:

- Source pane (left):** Contains R Markdown code. A brown box highlights the top section with the text: "Source pane : where the R Markdown file opens". The code includes a YAML header, a title, author, date, and output format, followed by R code chunks for setting up the environment, creating a cars object, and plotting pressure data.
- Environment pane (top right):** Shows the Global Environment. A green box highlights it with the text: "Environment pane: shows objects created by the code." The environment is currently empty.
- Console pane (bottom left):** Shows the R version (4.4.0) and copyright information. A red box highlights it with the text: "Console: shows commands, messages, and errors".
- Files/Plots pane (bottom right):** Shows the R Resources page. A blue box highlights it with the text: "Files/Plots panes: check saved outputs".

The R Markdown code in the Source pane is as follows:

```
1 ---
2 title: "Workshop_demo"
3 author: "Nancy Chi"
4 date: "2026-07-20"
5 output: html_document
6 ---
7
8 ```{r setup, include=FALSE}
9 knitr::opts_chunk$set(echo = TRUE)
10 ```
11
12 ## R Markdown
13
14 This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents.
15 For more details on using R Markdown see <http://rmarkdown.rstudio.com>.
16
17 When you click the **Knit** button a document will be generated that includes both content as well as the output of any
18 embedded R code chunks within the document. You can embed an R code chunk like this:
19
20 ```{r cars}
21 summary(cars)
22 ```
23
24 ## Including Plots
25
26 You can also embed plots, for example:
27
28 ```{r pressure, echo=FALSE}
29 plot(pressure)
30 ```
31
32 2:1 Workshop_demo.R R Markdown
```

The Console pane shows the following output:

```
R version 4.4.0 (2024-04-24) -- "Puppy Cup"
Copyright (C) 2024 The R Foundation for Statistical Computing
Platform: aarch64-apple-darwin20

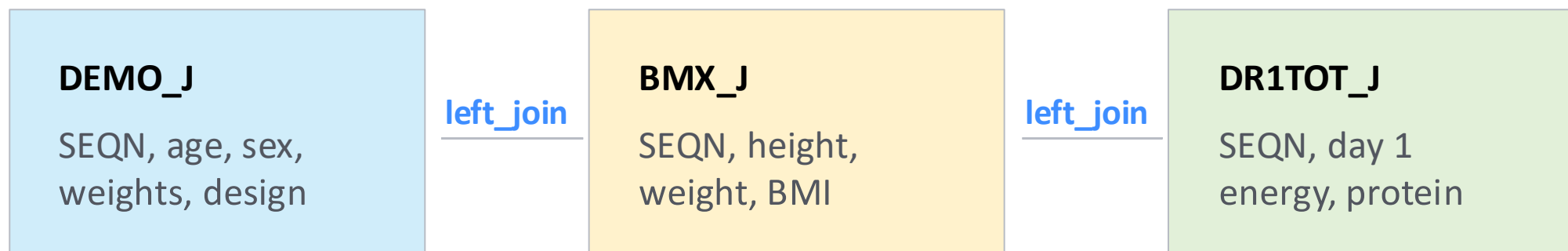
R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
```

The Files/Plots pane shows the R Resources page with links to R Resources, RStudio, Learning R Online, CRAN Task Views, R on StackOverflow, Getting Help with R, Posit Support, Posit Community Forum for the RStudio IDE, Posit Cheat Sheets, RStudio Packages, An Introduction to R, The R Language Definition, Writing R Extensions, R Installation and Administration, R Data Import/Export, R Internals, and Reference.

Most participant-level joins use SEQN as the respondent identifier



- Codebook

- SEQN - Respondent sequence number
- SDDSRVYR - Data release cycle
- RIDSTATR - Interview/Examination status
- RIAGENDR - Gender
- RIDAGEYR - Age in years at screening
- RIDAGEMN - Age in months at screening - 0 to 24 mos

Rule of thumb: inspect row counts and duplicate SEQN values before every join. In the demo, SEQN is the shared participant identifier across DEMO_J, BMX_J, and DR1TOT_J.

Joining tables and combining cycles are different operations

Join within a cycle (demo)

Use when columns live in different files for the same participants.

R verb: `left_join(..., by = "SEQN")`

Append across cycles

Use when rows come from different survey periods after harmonizing variables.

R verb: `bind_rows()`

For population estimates, pooled-cycle weights need explicit attention.



dplyr verbs you can map to Stata/Python habits

Task	R / dplyr	Stata mental model	Python / pandas
Keep variables	<code>select(SEQN, RIDAGEYR, BMXBMI)</code>	keep SEQN RIDAGEYR BMXBMI	<code>df[["SEQN", "RIDAGEYR", "BMXBMI"]]</code>
Filter rows	<code>filter(RIDAGEYR >= 20)</code>	keep if RIDAGEYR >= 20	<code>df[df["RIDAGEYR"] >= 20]</code>
Create variable	<code>Mutate(adult = RIDAGEYR >= 20)</code>	gen adult = RIDAGEYR >= 20	<code>df.assign(adult = df["RIDAGEYR"] >= 20)</code>
Merge files	<code>left_join(bmx, by = "SEQN")</code>	merge 1:1 SEQN using bmx	<code>df.merge(bmx, on = "SEQN", how = "left")</code>
Append cycles	<code>bind_rows(cycle1, cycle2)</code>	append using cycle2	<code>pd.concat([cycle1, cycle2], ignore_index = TRUE)</code>



A first ggplot should answer a data-checking question

Demo plot 1: BMI by age, colored by sex.

Demo plot 2: Day 1 energy intake by BMI group.

Save both figures with ggsave().

Ask before polishing

- Is the variable plausible?
- Are missing values visible?
- Are groups coded correctly?
- Does the plot match the research question?



Survey design matters before reporting estimates

For descriptive population estimates, NHANES analyses usually require the correct sample weight, strata, and PSU variables. Today we name these design elements; you may explore survey-weighted estimation on your own.

Weight

Which participants and measurement component does this represent?

Strata

Which design strata define variance estimation?

PSU

Which primary sampling units define clustering?



In-class activity: modify the BMI-by-age plot

- 01 Open the In-Class Practice section in the Rmd
- 02 Change color to BMI group
- 03 Add `facet_wrap(~ sex)`
- 04 Save `practice_bmi_by_age.png`
- 05 Write one pattern and one limitation



A credible NHANES analysis starts before modeling

Question

What population, cycle, and measurement domain?

Documentation

Which codebook, eligibility notes, and missing codes?

Assembly

Which joins by SEQN and which appends by cycle?

Design

Which weight, strata, and PSU for the estimate?

In NHANES, the quality of the final estimate depends as much on how we define, document, and assemble the data as it does on the model we choose.



Thank you

Demo handout: demo/nhanes_workshop_demo.html

R Markdown source: demo/nhanes_workshop_demo.Rmd

CDC data index: wwwn.cdc.gov/nchs/nhanes

